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Rapid Hubble constant inference from GW170817 using GPU-accelerated nested sampling: prior sensitivity and the limits of post-hoc reweighting

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ABSTRACT

The bright-siren measurement of the Hubble constant from GW170817 (Abbott et al. 2017b) relies on the assumption that switching from a volumetric luminosity-distance prior to a flat-in-redshift prior can be implemented by post-hoc reweighting of the baseline posterior samples, rather than by re-running the inference under the target prior. We test this assumption directly. Using a GPU-native heterodyned nested-sampling pipeline that completes the full $n_{\rm live} = 5000$ analysis in ~ 13 min on a single A100, we recompute the GW170817 H_0 posterior under four prior variants for the modern aligned-spin tidal waveform IMRPhenomXAS_NRTidalv3. Switching from the volumetric to a flat-in-redshift distance prior leaves the posterior mode essentially unchanged but inflates the high-tail probability $P(H_0 > 120\,{\rm km\,s^{-1}\,Mpc^{-1}})$ from 0.017 to 0.159 when the prior is imposed during sampling, while post-hoc reweighting of the baseline samples to the same target prior recovers only $P = 0.041$ – approximately 17 % of the prior-induced shift. The four nested-sampling evidences agree to $\Delta \ln Z \lesssim 1.8$, so the data do not prefer any one distance prior; the high-tail shift is therefore a property of the prior, not a data update. Targeted mode-isolated runs reveal an underlying (d_L, ι) bimodality with a high-H_0, low-d_L branch (Mode B; $H_0^{\rm MAP} \approx 110\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $|\ln \mathcal{B}_{\rm B/A}| < 1$ confirmed in two independent seeds) that the volumetric prior assigns negligible mass: this is the mechanism behind the reweighting deficit. The reweighted posterior has a lower effective sample size than the baseline ($n_{\rm eff}$ = 27,317 vs 30,695 from 1.78×10^5 baseline samples), independently flagging the coverage failure. The runtime budget makes full-sample prior-sensitivity reruns the right default robustness tool for bright-siren cosmology, in place of post-hoc reweighting.

Key words: gravitational waves – methods: data analysis – cosmological parameters – stars: neutron – software: data analysis

1 INTRODUCTION

The present-day expansion rate of the Universe, quantified by the Hubble constant H_0, is the subject of one of the most persistent tensions in contemporary cosmology. Measurements anchored in the early Universe, derived from cosmic-microwave-background observations assuming flat ΛCDM, favour a relatively low value (Planck Collaboration 2016), while late-Universe distance-ladder measurements based on Cepheids and Type Ia supernovae find a significantly higher value (Riess et al. 2016, 2022). The statistical and systematic discrepancy between these two families of measurement is now established at several σ, and its resolution – whether through unrecognised systematics or new cosmological physics – motivates independent probes of the expansion rate.

Gravitational-wave standard sirens are such a probe. The waveform amplitude of a compact-binary merger fixes the luminosity distance directly, without reference to an astrophysical distance ladder (Schutz 1986). When the merger is accompanied by an electromagnetic counterpart that identifies a host galaxy, the host redshift can be combined with the gravitational-wave distance to infer H_0 from a single event.

The binary neutron-star merger GW170817 (Abbott et al. 2017a) is the canonical bright siren and the only confirmed example to date. Its association with the kilonova AT2017gfo in NGC 4993 yielded the first gravitational-wave H_0 measurement, $H_0 = 70^{+12}_{-8}\,{\rm km\,s^{-1}\,Mpc^{-1}}$ at 68 % HPD (Abbott et al. 2017b).

For a single bright siren, the H_0 uncertainty is dominated by the luminosity-distance–inclination degeneracy: a face-on binary at larger d_L can produce the same observed strain as a more inclined binary at smaller d_L, with the two solutions mapping to different values of $H_0 \propto v_{\rm r}/d_L$ for a fixed host recessional velocity $v_{\rm r}$. The resulting posterior is broad and skewed, and only weakly informative on H_0 from one event. In such a regime, the measurement is more sensitive to prior choices than the statistical uncertainty alone would suggest, because the prior shapes the tails of the posterior rather than just its peak. The original GW170817 analysis adopted a volumetric distance prior $\pi(d_L) \propto d_L^2$ over $[10, 75]$ Mpc and reported that switching to a flat-in-redshift prior, implemented by *reweighting* posterior samples, produced only a minor change in the inferred H_0 posterior (Abbott et al. 2017b).

Reweighting is attractive because it avoids rerunning expensive Bayesian inference. For a posterior $p(\theta \mid d, \pi_0)$ obtained under baseline prior π_0, the target posterior under an alternative prior π_1

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can be estimated by assigning each sample a weight proportional to $\pi_1(\theta)/\pi_0(\theta)$. This procedure is statistically consistent provided the baseline samples cover the support of the target posterior. If π_1/π_0 is large in regions that the baseline run has sampled sparsely, the reweighted estimator has high variance and systematically low tail mass (see e.g. Speagle 2020). For GW170817, the baseline volumetric prior places little mass at low d_L; a flat-in-redshift target up-weights exactly this region, and the validity of reweighting is not obvious a priori.

Testing reweighting against a direct posterior obtained by sampling under the target prior requires repeated full gravitational-wave parameter estimation. In CPU frameworks such as BILBY (Ashton et al. 2019) and PARALLEL_BILBY, single-event binary-neutron-star analyses with phase-marginalised frequency-domain likelihood are sufficiently expensive that systematic exploration of distance, velocity, sampler, and waveform assumptions is rarely carried out in practice. Several recent developments make this constraint less severe: relative binning, also known as the heterodyned likelihood (Cornish 2013; Zackay et al. 2018), reduces the effective frequency grid for a long inspiral to a small number of bins; differentiable waveform libraries in JAX (Bradbury et al. 2018; Edwards et al. 2023) enable end-to-end GPU evaluation of the likelihood; and GPU-native nested-sampling kernels (Cabezas et al. 2024; Yallup et al. 2025; Prathaban et al. 2025) retain the evidence estimates that nested sampling provides for free while permitting large live-point counts and repeated full reruns. Related accelerated binary-neutron-star pipelines (Krishna et al. 2023; Wong et al. 2023; Wouters et al. 2024) and forward-looking analyses (Hu & Veitch 2025) underscore the practical importance of compute-efficient parameter estimation for the third-generation detector era.

In this paper we use a GPU-native nested-sampling pipeline, built on BLACKJAX-NS (Yallup et al. 2025; Prathaban et al. 2025) and a JAX implementation of the heterodyned likelihood, to revisit the GW170817 bright-siren H_0 inference of Abbott et al. (2017b). Our focus is not a new single-event measurement of H_0, but a controlled, multi-axis assessment of how that measurement depends on its priors, on the waveform, on the sampler hyperparameters, and on auxiliary inputs. We first validate the pipeline against GW150914 with the LVK GWTC-2.1+ production waveform IMRPhenomXPHM (Pratten et al. 2021; Abbott et al. 2024). We then report the IMRPhenomXAS_NRTidalv3 baseline GW170817 posterior, compare it with a direct flat-in-redshift posterior and with the reweighted flat-in-redshift posterior, and use targeted mode-isolated runs to identify the (d_L, ι) bimodality that mechanistically explains the reweighting deficit. A waveform cross-check against TaylorF2 bounds the residual waveform systematic. Robustness sweeps over sampler hyperparameters, heterodyne-bin count, PSD source, reference parameters, and host peculiar-velocity centre and width are referenced in Appendix A. We finish by reporting heterodyne speedup, live-point scaling, and a matched-pair sky-prior runtime sweep.

Section 2 describes the inference setup. Section 3 reports validation. Section 4 contains the prior-sensitivity analysis (the central science finding), the bimodality characterisation, and the waveform cross-check. Section 5 reports runtime and live-point scaling. Section 6 discusses implications and limitations. Section 7 concludes. Appendix A reports the inline robustness sweeps that are referenced in the body but not promoted to main figures.

2 METHOD

2.1 Nested sampling

For data d and source parameters θ under model $\mathcal{M}$, we target the posterior

$$p(\theta \mid d, \mathcal{M}) = \frac{\mathcal{L}(d \mid \theta, \mathcal{M})\, \pi(\theta \mid \mathcal{M})}{Z(d \mid \mathcal{M})}, \tag{1}$$

where $\mathcal{L}$ is the gravitational-wave likelihood, π the prior, and Z the Bayesian evidence. We estimate Z and draw posterior samples by nested sampling (Skilling 2006), using the GPU-native slice sampler implemented in BLACKJAX-NS (Yallup et al. 2025; Prathaban et al. 2025). The sampler evolves a population of live points through batched slice updates; this vectorisation keeps the GPU saturated when the per-evaluation likelihood cost is small.

All heterodyned GW170817 science runs reported here use $n_{\rm live} = 5000$, $n_{\rm delete} = 2500$ points removed per iteration ($\frac{1}{2}\, n_{\rm live}$), and $5\, n_{\rm dim}$ MCMC steps per slice; runs terminate when the fractional evidence increment is below 10^{-3}. Scaling experiments vary $n_{\rm live}$ while holding all other settings fixed. The GW150914 IMRPhenomXPHM validation run uses $n_{\rm live} = 8000$, $n_{\rm delete} = 2400$, $n_{\rm mcmc} = 160$ to verify that the validation conclusions are not sampler-limited at the smaller short-signal $n_{\rm dim}$.

2.2 Heterodyned likelihood

The full frequency-domain likelihood for GW170817 uses 259 201 bins over 20–2048 Hz at $\Delta f = 1/128$ Hz for the 128 s strain segment. We use a heterodyned (relative-binning) likelihood (Cornish 2013; Zackay et al. 2018), in which the ratio between a candidate waveform and a reference waveform is approximated by a piecewise-linear function over a small set of coarse frequency bins; we use 501 nominal bins for GW170817 and 383 for GW150914. The reference waveform parameters and the per-waveform effective bin count after frequency-support masking are recorded in each run's `config.json`. We compare heterodyned and unheterodyned posteriors directly in Section 3.2 to verify that the approximation does not bias the source-parameter posterior at the level needed for H_0 inference.

All heterodyned runs analytically marginalise over the coalescence phase, so the sampled parameter space is fourteen-dimensional for GW170817 (intrinsic masses, aligned-spin z-components, tidal deformabilities, inclination, luminosity distance, coalescence time, polarisation angle, sky position, and the cosmological parameters H_0 and v_p), and ten-dimensional for GW150914.

2.3 Joint Hubble-constant likelihood

We sample the Hubble constant jointly with the gravitational-wave source parameters rather than post-processing the d_L marginal. The GW170817 host galaxy NGC 4993 contributes a CMB-frame recessional velocity $v_{\rm r} = 3327\,{\rm km\,s^{-1}}$ with $\sigma_{v_{\rm r}} = 72\,{\rm km\,s^{-1}}$ and a Hubble-flow peculiar-velocity estimate $\langle v_p \rangle = 310\,{\rm km\,s^{-1}}$ with $\sigma_{v_p} = 150\,{\rm km\,s^{-1}}$. Both enter as Gaussian likelihood factors,

$$\ln \mathcal{L}_{\rm tot}(\theta, H_0, v_p) = \ln \mathcal{L}_{\rm GW}(d \mid \theta) + \ln \mathcal{N}\big(v_{\rm r}\,;\, H_0\, d_L + v_p,\, \sigma_{v_{\rm r}}\big) + \ln \mathcal{N}\Big(\langle v_p \rangle\,;\, v_p \tag{2}$$

following Abbott et al. (2017b).

2.4 Priors and prior variants

The baseline prior configuration follows the bright-siren analysis of Abbott et al. (2017b): $\pi(d_L) \propto d_L^2$ over $[10, 75]$ Mpc, $\pi(H_0) \propto H_0^{-1}$ over $[45, 250]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and $\pi(v_p) \propto 1$ over $[-1000, 1000]\,\mathrm{km\,s^{-1}}$ combined with the Gaussian v_p-likelihood term in equation 2 at $\sigma_{v_p} = 150\,\mathrm{km\,s^{-1}}$. Source-parameter priors use either a default-mass set (component masses uniform in $[0.5, 7.7]\,M_\odot$) or an LVK-bounds set (component masses uniform in $[0.87, 1.74]\,M_\odot$, matching Abbott et al. 2017b); both are full-sky, with right ascension uniform on $[0, 2\pi]$ and declination distributed according to a $\cos\delta$ prior on $[-\pi/2, \pi/2]$. Every heterodyned run reported here is full-sky; the optional ± 0.05 rad patch centred on NGC 4993 is used only in the matched-pair runtime study of Section 5.3.

We consider three prior variants in addition to the baseline:

(i) *Direct flat-in-redshift*: $\pi(z)$ uniform within the equivalent distance range, converted to a d_L prior via a flat ΛCDM cosmology and sampled directly.

(ii) *Reweighted flat-in-redshift*: the baseline posterior samples reweighted by the analytic ratio of the flat-in-redshift to volumetric d_L priors.

(iii) *Enlarged peculiar-velocity uncertainty*: identical to the baseline except $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$.

The direct and reweighted flat-in-redshift posteriors share the same target prior; they differ only in whether the prior is imposed during sampling or applied post-hoc.

2.5 Waveforms

For GW170817 we adopt two waveforms in the main analysis. The primary is IMRPhenomXAS_NRTidalv3, the most recent NR-calibrated tidal phase prescription on a modern aligned-spin base. TaylorF2 is included as a post-Newtonian inspiral-only family check. We additionally analysed IMRPhenomD_NRTidalv2 as an anchor waveform for the IMRX-vs-NRTidalv2 capture-fraction comparison reported inline in Section 4.2; the corresponding full posteriors are retained in the public test suite and summarised in Appendix A.

For GW150914 validation we use IMRPhenomXPHM (Pratten et al. 2021), the LVK GWTC-2.1+ production waveform for that event, which incorporates precession through a multi-scale-analysis treatment together with the dominant higher-order modes. Like-for-like comparison against the public LVK PE (Abbott et al. 2024; LIGO Scientific Collaboration et al. 2022) therefore requires no waveform substitution.

2.6 Posterior summaries

We summarise one-dimensional marginals by the maximum a posteriori (MAP) value and a sample-derived highest-posterior-density (HPD) interval at the requested credibility. The MAP is taken from the bin centre of the weighted histogram with maximum weighted count, on a fixed $1\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ grid over $[40, 230]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$; HPDs are the shortest contiguous interval containing the requested cumulative weight, computed directly from the weighted samples. We do not use kernel-density estimates for one-dimensional summaries because the GW170817 posterior tails carry the prior-sensitivity signal of Section 4.1 and KDE smoothing distorts the tails. KDEs are used only for two-dimensional corner contours, where the smoothing is geometrically justified.

Table 1. GW150914 validation: median values of the principal source parameters for our heterodyned IMRPhenomXPHM run, with a $n_{\mathrm{live}} = 5000$ cross-check. The LVK GWTC-2.1 IMRPhenomXPHM public PE (Abbott et al. 2024; LIGO Scientific Collaboration et al. 2022) reports $\mathcal{M}_c \approx 30.7\,M_\odot$, $q \approx 0.83$, $d_L \approx 440$ Mpc, $\iota \approx 2.62$ rad for direct comparison.

Run	$\mathcal{M}_c/M_\odot$	q	d_L/M
this work, IMRPhenomXPHM ($n_{\mathrm{live}} = 8000$)	30.35	0.87	455
this work, IMRPhenomXPHM ($n_{\mathrm{live}} = 5000$, cross-check)	30.34	0.87	460

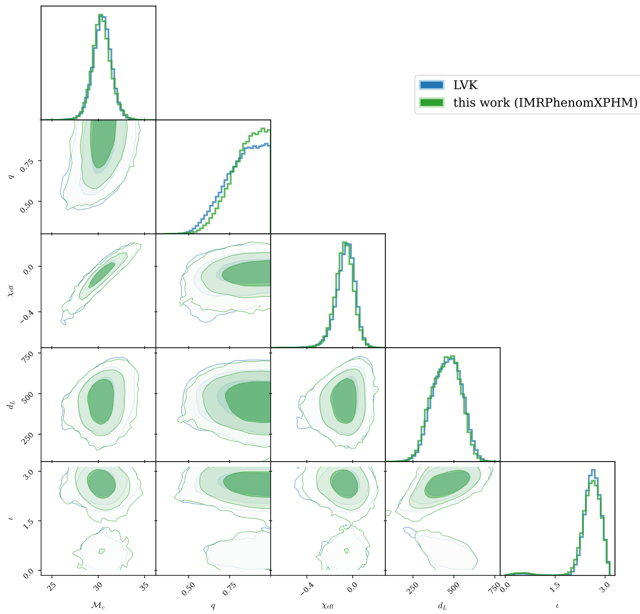

Figure 1. GW150914 validation: corner overlay of the LVK GWTC-2.1 IMRPhenomXPHM posterior (Abbott et al. 2024; LIGO Scientific Collaboration et al. 2022) and our heterodyned IMRPhenomXPHM run at $n_{\mathrm{live}} = 8000$, $n_{\mathrm{mcmc}} = 160$. Both panels show a like-for-like reproduction.

3 VALIDATION

3.1 GW150914

We validate the pipeline on the binary-black-hole event GW150914 (Abbott et al. 2016), which has a short signal and well-established public posteriors. We analyse the GWOSC strain data with IMRPhenomXPHM and the GWTC-2.1 PSDs, using the same heterodyned sampler as for GW170817 with phase marginalisation; for this validation run we increase the sampler settings to $n_{\mathrm{live}} = 8000$ with $n_{\mathrm{mcmc}} = 160$ slice steps per update, so that the comparison is not sampler-limited at the short-signal n_{dim}. Table 1 compares our recovered parameters with the LVK GWTC-2.1 public PE (Abbott et al. 2024; LIGO Scientific Collaboration et al. 2022); the corresponding corner overlay is shown in Figure 1. The chirp mass, mass ratio, luminosity distance, and inclination all agree with the LVK reference; in particular, $d_L^{\mathrm{med}} = 455$ Mpc for our IMRPhenomXPHM run is within $\sim$15 Mpc of the LVK value. The validation is therefore a like-for-like LVK reproduction.

3.2 Heterodyned vs unheterodyned consistency on GW170817

We next verify that the heterodyned approximation does not bias the GW170817 source-parameter posterior. We run the same sam-

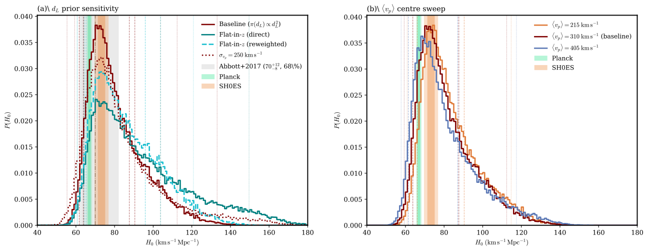

Figure 2. Prior-sensitivity comparison for IMRPhenomXAS_NRTidalv3. *Left panel (a):* weighted step-histograms of H_0 under the four distance-prior variants of Section 2.4 (volumetric baseline, direct flat-in-z, reweighted flat-in-z, and $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$). The directly sampled flat-in-redshift posterior places substantially more mass at high H_0 than the reweighted estimate, despite both targeting the same prior. *Right panel (b):* peculiar-velocity centre sweep — H_0 posteriors for $\langle v_p \rangle \in \{215, 310, 405\}\,\mathrm{km\,s^{-1}}$ at fixed $\sigma_{v_p} = 150\,\mathrm{km\,s^{-1}}$ (s18 runs). The published Abbott+2017 $H_0 = 70^{+12}_{-8}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ band (Abbott et al. 2017b) is shown in panel (a) as a vertical reference. Planck ΛCDM and SH0ES distance-ladder bands are shown in both panels.

pler on the full 259 201-bin frequency-domain likelihood for both IMRPhenomD_NRTidalv2 and TaylorF2 at $n_{\rm live} = 1500$, and additionally at $n_{\rm live} \in \{500, 2500\}$ for IMRPhenomD_NRTidalv2. The heterodyned and unheterodyned d_L, ι, and H_0 marginals overlay closely, and the heterodyned $H_0^{\rm MAP}$ lies between the surrounding unheterodyned cross-checks at the level of the sampler-statistical width. We conclude that the heterodyned approximation is adequate for H_0 inference at this event's signal-to-noise ratio; the corresponding wall-clock reduction is quantified in Section 5.2.

4 RESULTS

4.1 Prior sensitivity (the central science finding)

The central scientific result of this paper is a direct comparison between the flat-in-redshift H_0 posterior obtained by sampling under the target prior and that obtained by reweighting the baseline posterior to the same target prior, under the locked primary waveform IMRPhenomXAS_NRTidalv3. We carry out the four prior variants of Section 2.4 on the default-mass full-sky prior set and report sample-derived HPDs in Table 2; the corresponding marginals are shown in Figure 2.

4.1.0.1 Direct-vs-reweighted comparison. The direct flat-in-redshift run leaves the posterior MAP essentially unchanged at $70.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ but materially broadens the 68 per cent HPD on the high side from 87.6 to $103.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, and raises the high-tail probability

$$P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}}) = 0.017 \quad \text{(baseline)} \quad \longrightarrow \quad 0.159 \quad \text{(direct flat-in-z)}$$

together with $P(H_0 > 150\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ from $< 10^{-4}$ to 0.038. The reweighted estimate captures only a small fraction of this shift: $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}}) = 0.041$, equivalent to

$$\frac{0.041 - 0.017}{0.159 - 0.017} \approx 17\,\% \tag{3}$$

of the direct prior-induced change. The 95 per cent upper bound under reweighting only reaches $95.9\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, far short of the directly sampled $103.8\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ at 68 per cent. The four IMRPhenomXAS_NRTidalv3 evidence values agree to $\Delta \ln Z \lesssim 1.8$ – not decisive on the Jeffreys scale, and within the run-to-run variability seen in the public repository for this configuration. The data therefore do not prefer one distance prior over another at any meaningful significance: the change in the high-H_0 tail across the prior variants is a *prior* effect, not a data-driven update.

4.1.0.2 Effective-sample-size diagnostic. The reweighting failure is independently flagged by the effective sample size of the reweighted posterior. From the same 1.78×10^5-sample IMRPhenomXAS_NRTidalv3 baseline we measure $n_{\rm eff} = 30{,}695$ (efficiency 17.3 %); the directly sampled flat-in-redshift run has $n_{\rm eff} = 37{,}022$ (18.4 %), while the reweighted flat-in-redshift estimator has $n_{\rm eff} = 27{,}317$ – *lower* than the baseline despite reweighting from the same draw. This is the conventional symptom of importance-sampling weights concentrated on a small subset of the baseline draw, exactly the regime in which reweighting is known to be unreliable. The diagnostic is cheap, requires no additional GPU time, and would have flagged the deficit in the original analysis from the baseline draw alone.

4.1.0.3 Peculiar-velocity sweep. The $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ variant shifts the H_0 posterior only marginally: $P(H_0 > 120)$ moves from 0.017 to 0.069. Sweeping the v_p prior *centre* over the historical literature range $\langle v_p \rangle \in \{215, 310, 405\}\,\mathrm{km\,s^{-1}}$ at fixed $\sigma_{v_p} = 150\,\mathrm{km\,s^{-1}}$ (Figure 2b, Table 2) shifts $H_0^{\rm MAP}$ by $6\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ peak-to-peak (74.5 at $\langle v_p \rangle = 215\,\mathrm{km\,s^{-1}}$ to 68.5 at $\langle v_p \rangle = 405\,\mathrm{km\,s^{-1}}$) while $P(H_0 > 120)$ changes by less than 0.02. The $\langle v_p \rangle = 310\,\mathrm{km\,s^{-1}}$ run replicates the IMRPhenomXAS_NRTidalv3 baseline to within 0.01 in $\ln Z$. For GW170817 with NGC 4993 identified, the dominant prior-induced uncertainty in H_0 is therefore the distance prior, not the peculiar-velocity prior.

4.1.0.4 Cross-waveform check. Repeating the four-variant prior sweep on the legacy aligned-spin tidal model IMRPhenomD_NRTidalv2 (NRTidalv2 calibration) gives qualitatively the same behaviour with a larger amplitude: baseline $P(H_0 > 120) = 0.076$; flat-in-z direct $P = 0.281$; reweighted $P = 0.195$; reweighting capture fraction $(0.195 - 0.076)/(0.281 - 0.076) \approx 58\,\%$. The newer NRTidalv3 calibration in IMRPhenomXAS_NRTidalv3 tightens the upper H_0 tail relative to NRTidalv2, leaving less Mode-B mass for the prior shift to redistribute and therefore both reducing the absolute prior-induced change in $P(H_0 > 120)$ *and* reducing the fraction of that change that reweighting can recover. The qualitative conclusion – that reweighting systematically understates the prior-induced change in the high-H_0 tail – is therefore robust across the tidal-calibration axis, and is in fact *more* severe under the modern waveform than under the older one.

We caution against over-interpreting these shifts as physical results. None of the prior variants changes the qualitative statement that GW170817 alone is consistent with both early- and late-Universe H_0 measurements. The methodological point is that the magnitude of the shift under a change of distance prior is a property of the inference rather than of the cosmology, and that the magnitude one would report using reweighting alone is materially different from the magnitude recovered by direct sampling.

4.2 The d_L–ι bimodality

The flat-in-redshift posterior is broad and asymmetric, with non-negligible weight at $d_L \lesssim 25\,\mathrm{Mpc}$. To resolve whether this is a

Table 2. IMRPhenomXAS_NRTidalv3 H_0 summaries for the four prior variants on the default-mass full-sky prior set, $n_{\rm live} = 5000$. MAP and sample-derived 68 per cent HPDs in km s^{-1} Mpc^{-1}. The reweighted variant is post-hoc and has no independent $\ln Z$. All numbers generated by `Plots/build_paper_tables.py` from the canonical sample CSVs.

Variant	MAP	68% HPD	$P(H_0 > 120)$	$P(H_0 > 150)$	$\ln Z$
Baseline ($\pi(d_L) \propto d_L^2$)	70.5	[63.8, 87.6]	0.017	$< 10^{-4}$	486.25 ± 0.11
Flat-in-z, direct	70.5	[64.2, 103.8]	0.159	0.038	487.30 ± 0.10
Flat-in-z, reweighted	73.5	[65.2, 95.9]	0.041	0.000	(post-hoc)
$\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$	73.5	[61.7, 90.5]	0.069	0.015	485.55 ± 0.09
$\langle v_p \rangle = 215\,\mathrm{km\,s^{-1}}$	74.5	[66.1, 90.4]	0.021	$< 10^{-4}$	485.76 ± 0.12
$\langle v_p \rangle = 405\,\mathrm{km\,s^{-1}}$	68.5	[61.5, 87.0]	0.030	0.000	486.35 ± 0.10

single broadened mode or a genuine bimodality, we run two prior-restricted variants of the direct flat-in-redshift analysis with IMRPhenomD_NRTidalv2 at $n_{\rm live} = 5000$: a Mode-A run with $d_L \in [30, 75]$ Mpc and a Mode-B run with $d_L \in [10, 30]$ Mpc. We retain the original GWTC-1 heterodyne reference, and additionally run an unrestricted $d_L \in [10, 75]$ Mpc flat-in-redshift analysis with the heterodyne reference re-anchored at Mode B ($d_L = 20$ Mpc, $\iota = 2.0$ rad) to test whether the apparent Mode-B mass is a heterodyne-reference artefact.

Figure 3 shows the joint (d_L, ι) posterior under the unrestricted run, together with the per-mode H_0 marginals. The support is a curved arm running from $(d_L, \iota) \approx (15, 1.9)$ Mpc·rad to $(40, 2.7)$ Mpc·rad with two distinct local maxima. Table 3 reports the local evidences and per-mode H_0 summaries.

The Bayes factor between Mode B and Mode A, after correcting for the volume difference between the restricted priors, is

$$\ln \mathcal{B}_{\rm B/A} = (\ln Z_{\rm B} - \ln Z_{\rm A}) + \ln(20/45) = +0.15 - 0.81 = -0.66 \quad \text{(seed=0)}. \tag{4}$$

An independent second-seed replication (Table 3, lower block) gives $+0.91 - 0.81 = +0.10$ (seed=1). Both seeds satisfy $|\ln \mathcal{B}_{\rm B/A}| < 1$, placing the result in the "not worth more than a mention" range of the Jeffreys scale; the sign difference between seeds is consistent with the ~ 0.1 per-run $\ln Z$ uncertainty. Mode B is therefore neither significantly favoured nor disfavoured by the data. It contributes a non-negligible share of the joint posterior under any prior that gives the low-d_L region appreciable weight. The Mode-B-anchored unrestricted run recovers $P(H_0 > 120) = 0.281$ – statistically indistinguishable from the GWTC-1-anchored direct flat-in-redshift run reported in Section 4.1 – so the heterodyne-reference choice does not bias the Mode-A/Mode-B weight ratio.

This is the mechanism behind the reweighting capture fraction of equation 3. Mode B is the high-H_0, small-d_L branch of the distance–inclination degeneracy. The baseline volumetric prior $\pi(d_L) \propto d_L^2$ assigns Mode B a tiny prior mass (the [10, 30] Mpc slab carries less than 5 % of the total [10, 75] Mpc volumetric prior), and the corresponding region of the baseline posterior is sparsely sampled. Reweighting cannot reconstruct what is not there. Direct sampling under the flat-in-redshift prior populates Mode B at sampler resolution, and it is precisely the missing Mode-B mass that opens the gap between the directly sampled and reweighted estimates of $P(H_0 > 120)$. Posterior-coverage diagnostics that compare the baseline samples against the target prior – such as the $n_{\rm eff}$ measurement reported in Section 4.1 – should therefore be standard practice before reporting a reweighted bright-siren H_0 summary.

4.3 Cross-waveform Hubble-constant posterior

Figure 4 shows the GW170817 H_0 posterior under the LVK-bounds baseline prior for IMRPhenomXAS_NRTidalv3 and TaylorF2, together with the published Abbott+2017 GW170817 H_0 band as a vertical reference. All curves are weighted step-histograms; sample-derived HPDs are reported in Table 4. Figure 5 shows the corresponding $(\mathcal{M}_c, q, \chi_{\rm eff}, d_L, \iota)$ corner overlay against the public GWTC-1 IMRPhenomPv2_NRTidal posterior (Abbott et al. 2019; LIGO Scientific Collaboration & Virgo Collaboration 2018).

IMRPhenomXAS_NRTidalv3 has $H_0^{\rm MAP} = 71.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ and 68 per cent HPD $[63.1, 88.5]\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. TaylorF2 sits $\sim 2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ lower in MAP – a small offset, expected for a lower-PN, point-particle inspiral family – and overlaps IMRPhenomXAS_NRTidalv3 within their 68 per cent HPDs. The published Abbott+2017 result $H_0 = 70^{+12}_{-8}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ (Abbott et al. 2017b) is consistent with both rows. IMRPhenomXAS_NRTidalv3 has the tighter upper tail (95 per cent upper bound 117.6 versus $168.2\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ for TaylorF2), which we attribute to the more recent NR-calibrated tides damping the high-H_0, small-d_L branch of the degeneracy. The IMRX-vs-IMR capture-fraction comparison reported in Section 4.1 indicates that the same tide-driven tail compression accounts for the smaller absolute prior-induced shift in $P(H_0 > 120)$ under IMRPhenomXAS_NRTidalv3.

5 PERFORMANCE AND RUNTIME STUDIES

5.1 Wall-clock runtime

The primary heterodyned GW170817 H_0 analysis with IMRPhenomXAS_NRTidalv3 and $n_{\rm live} = 5000$ on the LVK-bounds prior set completes in ≈ 13 min on a single NVIDIA A100 GPU; the corresponding TaylorF2 run takes ≈ 9 min. The full four-variant prior-sensitivity suite (baseline, direct flat-in-z, reweighted post-hoc, $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$) for IMRPhenomXAS_NRTidalv3 on the default-mass prior set fits inside an hour. The GW150914 IMRPhenomXPHM validation run at the larger $n_{\rm live} = 8000$, $n_{\rm mcmc} = 160$ settings takes ≈ 5 h on the same hardware – the price for both the precessing waveform and the converged short-signal sampler configuration that yields the like-for-like LVK reproduction of Section 3.1.

5.2 Heterodyne speedup and live-point scaling

The heterodyned likelihood is the enabling approximation. Figure 6 compares heterodyned and unheterodyned IMRPhenomD_NRTidalv2 wall-clock time on GW170817 across $n_{\rm live} \in \{500, 1000, 2500, 5000, 10\,000, 20\,000, 50\,000, 100\,000\}$ for the heterodyned sampler and $n_{\rm live} \in \{500, 1500, 2500\}$ for the unhetero-

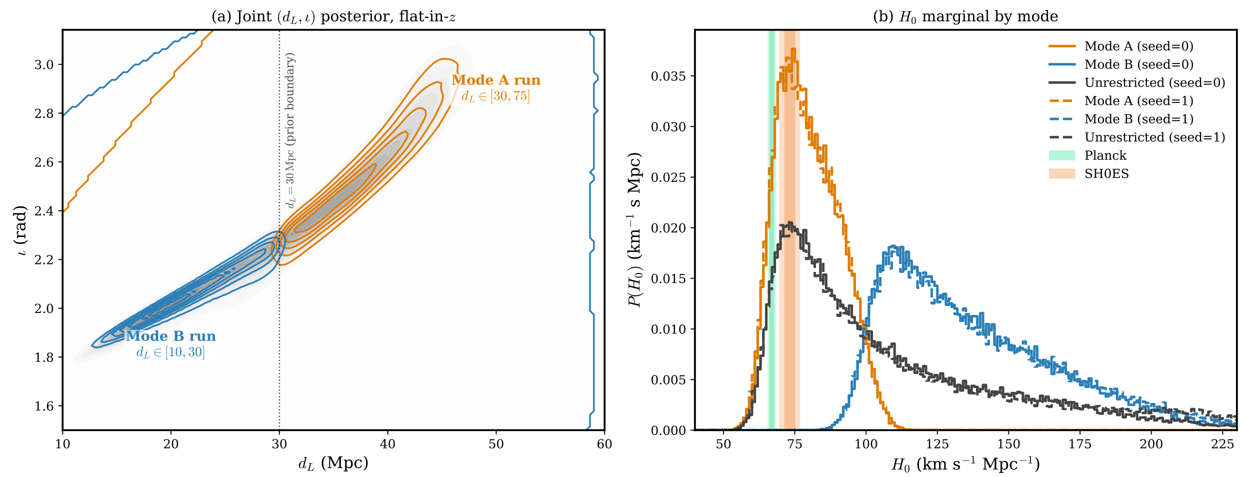

Figure 3. The d_L–ι bimodality in the GW170817 flat-in-redshift posterior. Left: joint (d_L, ι) posterior from the unrestricted $d_L \in [10, 75]$ Mpc run (Mode-B-anchored heterodyne reference); the prior-restricted Mode-A and Mode-B contours from the two separate restricted-prior runs are overlaid in distinct colours, with the $d_L = 30$ Mpc prior boundary marked. Right: per-mode H_0 marginals as weighted step-histograms. The Planck ΛCDM and SH0ES distance-ladder bands are shown for context.

Table 3. Local evidences, MAP H_0 values, and sample-derived 68 per cent HPDs for the d_L-restricted flat-in-redshift runs. Upper block: seed=0 (primary); lower block: independent seed=1 replication (s18 runs). The Mode-B/Mode-A Bayes factor includes a $\ln(20/45)$ correction for the difference in restricted-prior d_L volumes.

Variant	d_L range (Mpc)	$H_0^{\rm MAP}$ (km s^{-1} Mpc^{-1})	68% HPD	$P(H_0 > 120)$	$\ln Z$
Mode A	[30, 75]	74.5	[66.7, 88.8]	$< 10^{-4}$	486.80 ± 0.10
Mode B	[10, 30]	109.5	[99.5, 151.8]	0.638	486.95 ± 0.09
Unrestricted	[10, 75]	73.5	[62.6, 116.5]	0.281	486.48 ± 0.09
Mode A (s=1)	[30, 75]	72.5	[66.1, 88.6]	$< 10^{-4}$	486.71 ± 0.09
Mode B (s=1)	[10, 30]	110.5	[99.0, 154.5]	0.646	487.62 ± 0.10
Unrestr. (s=1)	[10, 75]	73.5	[62.0, 121.1]	0.311	487.52 ± 0.09

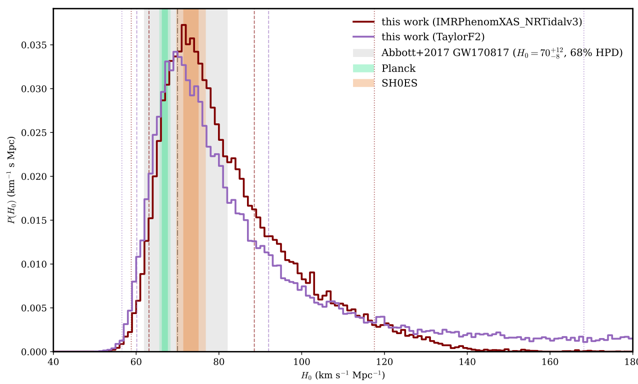

Figure 4. GW170817 H_0 posterior for IMRPhenomXAS_NRTidalv3 (primary, LVK-bounds baseline) and TaylorF2 (family check, LVK-bounds baseline). Vertical markers are 68 per cent and 95 per cent HPD bounds derived from the weighted samples (no KDE smoothing); sample-derived HPDs are reported in Table 4. The Planck ΛCDM and SH0ES distance-ladder values are shown for context. The published Abbott+2017 GW170817 $H_0 = 70^{+12}_{-8}$ km s^{-1} Mpc^{-1} band (Abbott et al. 2017b) is shown as a vertical reference.

dyned reference, plus the TaylorF2 unheterodyned point at $n_{\rm live} =$

Table 4. GW170817 H_0 summaries for the two-waveform main set on the LVK-bounds baseline prior, $n_{\rm live} = 5000$. MAP and sample-derived HPDs in km s^{-1} Mpc^{-1}.

Waveform	MAP	68% HPD	95% HPD	P
IMRPhenomXAS_NRTidalv3 (primary)	71.5	[63.1, 88.5]	[58.8, 117.6]	
TaylorF2 (family check)	69.5	[60.2, 92.0]	[56.5, 168.2]	

1500. At matched $n_{\rm live}$ the speedup factors are

$$\text{speedup}(n_{\rm live}) = \begin{cases} 31.1\times & n_{\rm live} = 500, \\ 50.7\times & n_{\rm live} = 1500, \\ 67.7\times & n_{\rm live} = 2500. \end{cases} \quad (5)$$

The relative-binning approximation accounts for the bulk of the wall-clock reduction; the GPU saturation provided by BLACKJAX-NS is layered on top.

The heterodyned $H_0^{\rm MAP}$ is stable across $n_{\rm live} \geq 2500$ to within 1 km s^{-1} Mpc^{-1}, and $\ln Z$ stabilises within ±0.2 over the same range. The $n_{\rm live} = 5000$ configuration adopted for the science runs is therefore in the converged regime. Live-point counts up to 10^5 remain accessible on a single GPU; the $n_{\rm live} = 10^5$ heterodyned run completes in roughly four hours.

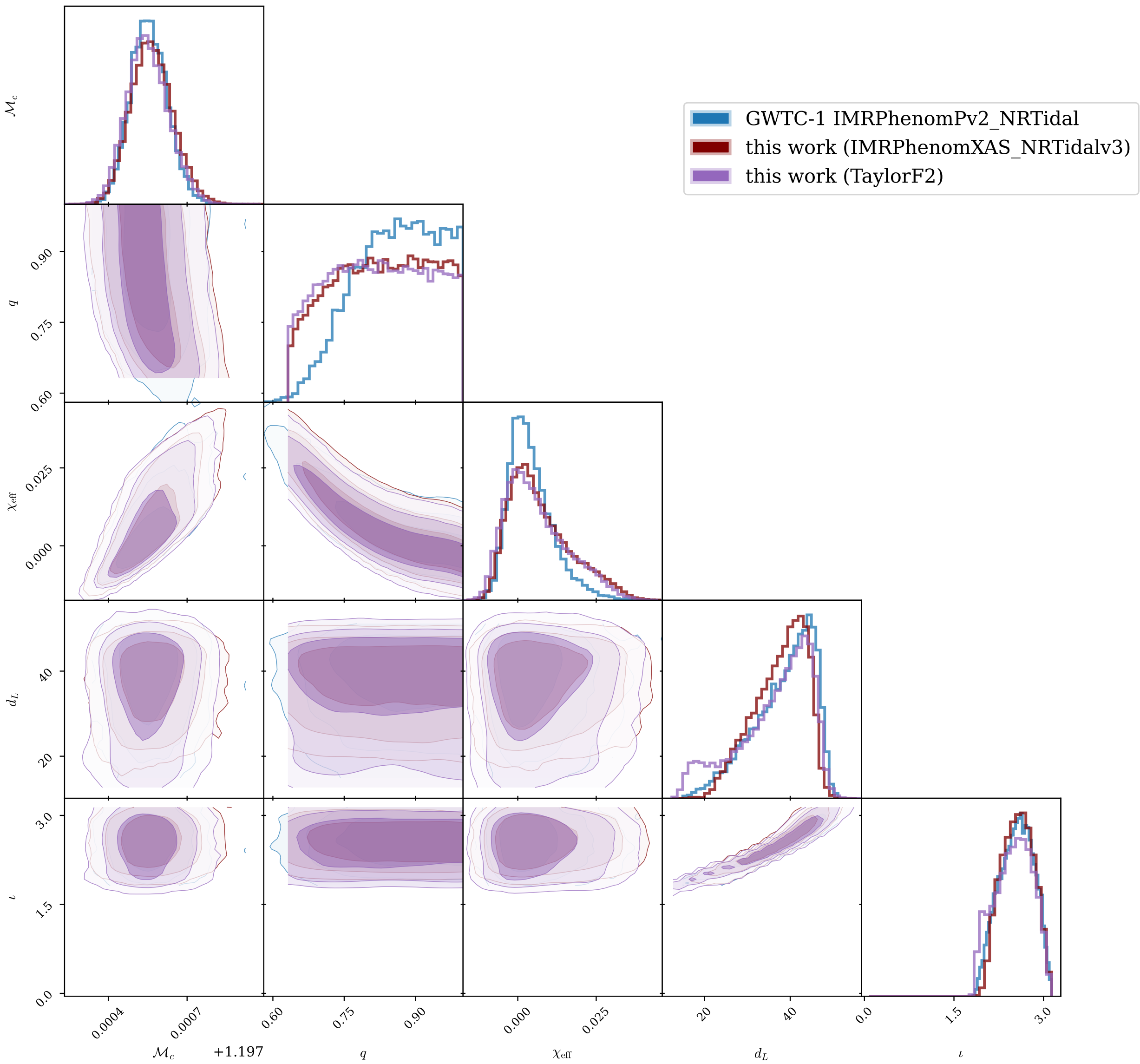

Figure 5. GW170817 source-parameter corner overlay for IMRPhenomXAS_NRTidalv3 and TaylorF2 on the LVK-bounds baseline prior, with the public GWTC-1 IMRPhenomPv2_NRTidal posterior (Abbott et al. 2019; LIGO Scientific Collaboration & Virgo Collaboration 2018) included as a literature reference. Two-dimensional contours are KDE-smoothed; one-dimensional marginals are weighted step-histograms.

5.3 Sky-prior matched-pair runtime sweep

We measured the wall-clock cost of imposing a tight ±0.05 rad patch around NGC 4993 on the heterodyned sampler at the science configuration $n_{\rm live}$ = 5000, using IMRPhenomD_NRTidalv2 for a matched-pair comparison against the corresponding s07 full-sky run: the narrow-sky run takes 735 s versus 858 s full-sky, a wall-clock ratio of 0.86. Two unheterodyned matched pairs at the smaller live-point count $n_{\rm live}$ = 1500 – where unheterodyned runs are computationally tractable – give a host-localised/full-sky ratio of 0.98 for IMRPhenomD_NRTidalv2 (20,042 s host-localised vs 20,474 s full-sky) and 1.44 for TaylorF2 (8615 s host-localised vs 5981 s full-sky). The sky-prior tightness therefore changes wall-clock by less than a factor of two in either direction, and is not the source of the heterodyne speedup. We caution that the unheterodyned pairs use the historical "host-localised" prior of the GWTC-1 reference scripts, which is wider than the s15 ±0.05 rad patch; the conclusion – that sky-prior tightness is not the dominant runtime axis – is robust across the two definitions but the precise ratios are not directly comparable.

5.4 Scope of the performance claim

We deliberately do not compare runtime against a single-core CPU baseline, which would give a misleadingly large speedup factor, nor against a production PARALLEL_BILBY run that we have not executed under matched settings. A like-for-like GW170817 PARALLEL_BILBY benchmark using IMRPhenomXAS_NRTidalv3 with identical priors and live-point count is the appropriate CPU comparison and is left for future work. The claim we make is narrower: on a single A100 GPU, the full $n_{\rm live}$ = 5000 H_0 analysis completes in under a quarter of an hour, the multi-variant prior-sensitivity suite fits inside an hour, the live-point convergence study extends to $n_{\rm live}$ = 10^5 within an overnight budget, and the sky-prior matched-pair sweep takes well

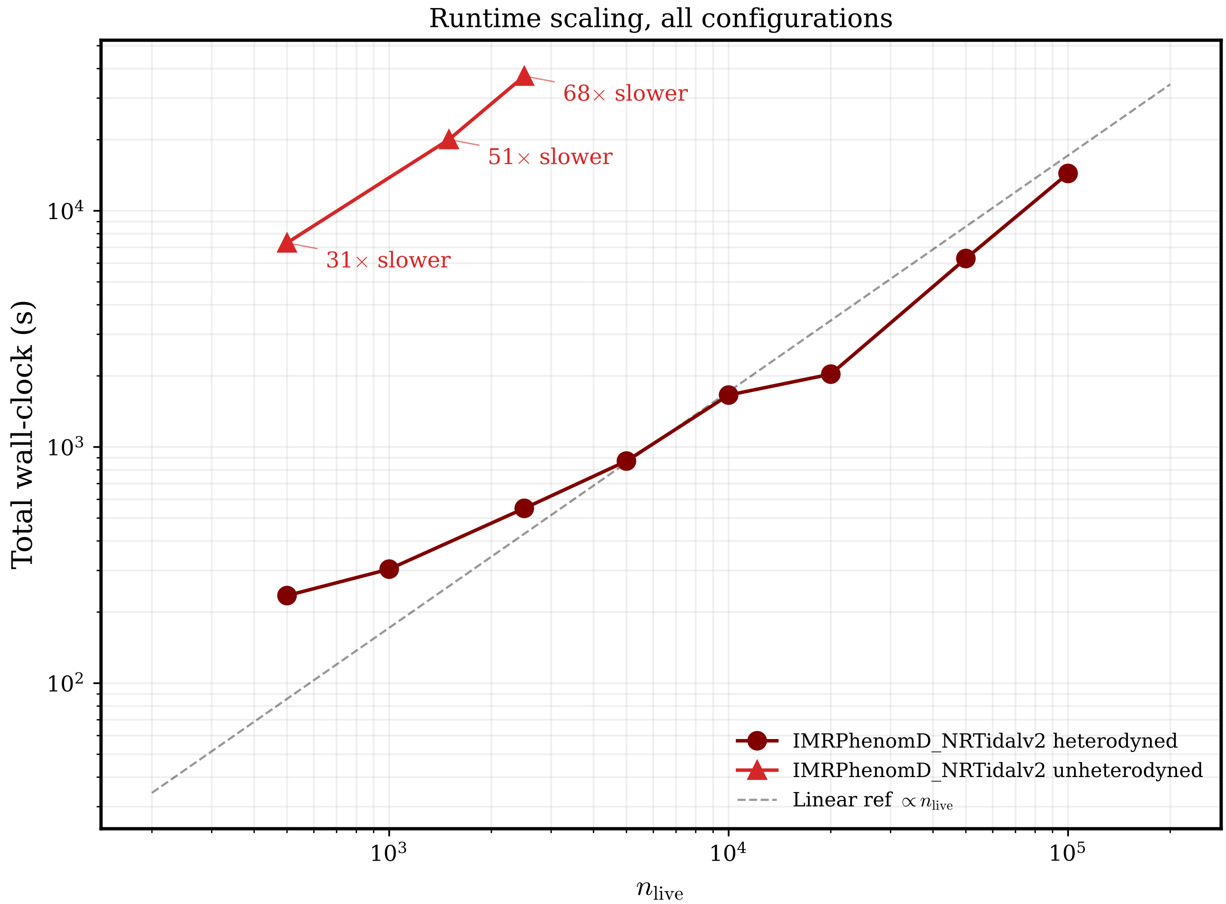

Figure 6. GW170817 wall-clock scaling. Heterodyned and unheterodyned wall-clock as a function of $n_{\rm live}$ for the IMRPhenomD_NRTidalv2 family on the default-mass full-sky prior set, with the TaylorF2 unheterodyned point at $n_{\rm live} = 1500$ for context. The heterodyned series has log–log gradient $\simeq 0.60$ over the plotted range; matched unheterodyned points are 31×/51×/68× slower at $n_{\rm live} = 500/1500/2500$. The shallow bend at $n_{\rm live} = 20\,000$ is a termination-tolerance artefact and is documented in the public repository.

under a working day. This runtime budget is what makes the multi-axis robustness study scientifically routine rather than exceptional.

6 DISCUSSION

6.1 Implications for bright-siren cosmology

The main methodological message is that posterior reweighting, although statistically consistent in the limit of infinite baseline samples, can substantially understate the prior sensitivity of bright-siren H_0 measurements in practice. For GW170817 under IMRPhenomXAS_NRTidalv3, switching from a volumetric distance prior to a flat-in-redshift prior changes $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ by a factor of ≈ 9 under direct sampling, while reweighting captures only ≈ 17 % of that shift (equation 3). The corresponding capture fraction on IMRPhenomD_NRTidalv2 – where the high-H_0 tail is heavier – rises to ≈ 58 %. The deficit is a problem of effective sample coverage rather than of statistical methodology: the baseline volumetric prior assigns Mode B a small prior mass and reweighting cannot reconstruct what was not sampled. The reweighted posterior's effective sample size, lower than the baseline's despite reweighting from the same draw, independently flags the coverage failure and is the diagnostic we recommend as a default check before any reweighted bright-siren H_0 summary is reported.

For a single event this under-estimation does not change the cosmological interpretation: the GW170817 posterior is broad enough that both early- and late-Universe H_0 values lie within its 68 per cent HPD under any of the priors considered here. However, the scale of the effect matters for forecasts and combined analyses. When bright-siren posteriors are combined over multiple events, or when tail probabilities are compared against external priors from cosmological surveys, the prior-dependence of each single-event posterior propagates through. An analysis that quantifies that dependence only through reweighting risks a systematic bias in the direction of under-estimating the prior contribution.

6.2 Where reweighting is sufficient

Reweighting is adequate when the target and baseline priors have similar support in the regions that carry most of the posterior mass. The $\sigma_{v_p} = 250\,\mathrm{km\,s^{-1}}$ variant in our analysis is a case of this kind: the change modifies the host-velocity likelihood smoothly over the scale of the velocity posterior, the baseline samples cover the affected region well, and the reweighted and directly sampled σ_{v_p} posteriors agree to well within the statistical width. The flat-in-redshift case fails the same coverage test, because the baseline volumetric prior and the target flat-in-redshift prior differ most at the edges of the d_L range. The mode-isolated runs of Section 4.2 make this concrete:

Mode B has $|\ln \mathcal{B}_{B/A}| < 1$ relative to Mode A (confirmed over two independent seeds) under the flat-in-redshift prior, but its baseline-volumetric-prior mass is below 5 % of Mode A's, so reweighted samples are essentially blind to it.

6.3 Implications for future bright sirens

The practical consequence is that direct prior sampling should be used for bright-siren H_0 analyses whenever the target prior differs non-trivially in support from the baseline, rather than being used only as a cross-check. The runtime budget on a single A100 GPU now permits such reruns routinely (Section 5); the live-point and sampler-robustness studies (Appendix A) show that the science-quality configuration sits in the converged regime, so the marginal cost of a prior-sensitivity rerun is the marginal cost of one ∼15 min run. Live-point counts up to 10^5 are accessible on the same hardware, which means that prior-sensitivity studies can in principle be combined with population-level analyses without a substantial increase in end-to-end wall-clock time. This is directly relevant to the third-generation detector era (Hu & Veitch 2025), where the detection rate of compact-binary mergers will make repeated, controlled posterior reanalyses the norm rather than the exception.

6.4 Limitations and open items

Several limitations should be acknowledged. First, no waveform in our JAX inventory simultaneously includes precession and tides; the locked primary is therefore the aligned-spin tidal IMRPhenomXAS_NRTidalv3. Second, the host-galaxy association with NGC 4993 and the peculiar-velocity inputs are taken directly from Abbott et al. (2017b); we have varied σ_{v_p} and $\langle v_p \rangle$ over the historical literature range (Appendix A). Third, we do not present a matched PARALLEL_BILBY benchmark, so the performance claim is restricted to single-A100 wall-clock runtime on the current pipeline. Fourth, the d_L–ι bimodality is characterised at the level of local evidences and mode-isolated marginals; we do not at present propagate Mode B into a population analysis or combine it with an electromagnetic likelihood for the host inclination.

7 CONCLUSIONS

We have used a GPU-native nested-sampling pipeline to revisit the GW170817 bright-siren H_0 measurement of Abbott et al. (2017b). Switching the distance prior from volumetric to flat-in-redshift by direct sampling raises $P(H_0 > 120\,\mathrm{km\,s^{-1}\,Mpc^{-1}})$ from 0.017 to 0.159, while post-hoc reweighting captures only 0.041, ≈ 17 % of the shift. The four IMRPhenomXAS_NRTidalv3 evidence values agree to $\Delta \ln Z \lesssim 1.8$, so the high-tail shift is a property of the prior rather than a data update. The mechanism is the d_L–ι bimodality: a high-H_0, small-d_L branch (Mode B; $|\ln \mathcal{B}_{B/A}| < 1$, confirmed in two independent seeds) carries appreciable likelihood but negligible volumetric-prior mass, and reweighting cannot recover what was not sampled. The reweighted estimator's effective sample size, lower than the baseline's, independently flags the coverage failure. The runtime budget makes direct prior-sensitivity reruns the right default for bright-siren cosmology in place of post-hoc reweighting, with multi-axis robustness studies now routine on a single A100 GPU.

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TODO ASK WILL: supervisor, research-group, and compute-resource (Google Cloud A100) acknowledgements; funding acknowledgements. We acknowledge GWOSC and the LIGO–Virgo–KAGRA Collaboration for the public strain data and reference posteriors used here (Abbott et al. 2019, 2024; LIGO Scientific Collaboration & Virgo Collaboration 2018, 2017; LIGO Scientific Collaboration et al. 2022).

DATA AVAILABILITY

This paper uses publicly available GW170817 and GW150914 strain data and GWTC-1/GWTC-2.1 reference posteriors from the Gravitational Wave Open Science Center (GWOSC) and from LIGO Scientific Collaboration & Virgo Collaboration (2018, 2017); LIGO Scientific Collaboration et al. (2022). All derived posterior-sample CSV files, summary tables, scaling logs, and figure-generation scripts are distributed with the project repository at `TODO:insertpermanentrepositoryURL/DOI`, with provenance recorded in `paper_knowledge_base/result_link_index.md` and the scaling-study summary CSV. Numerical entries in Tables 1, 4, 5, and 6 are regenerated from the canonical sample CSVs by `Plots/build_paper_tables.py`.

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APPENDIX A: INLINE ROBUSTNESS SWEEPS

We complete the multi-axis robustness picture with sweeps that we ran but do not display as primary figures, since the central science finding of Section 4.1 is unaffected by any of them.

- *Sampler hyperparameters.* On the IMRPhenomD_NRTidalv2 default-mass baseline at $n_{\rm live} = 5000$, varying $n_{\rm delete}/n_{\rm live} \in \{0.10, 0.25, 0.50, 0.75\}$ moves $H_0^{\rm MAP}$ by $\leq 1.2\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $\ln Z$ by ≤ 0.6; varying the heterodyne-bin count $n_{\rm bins} \in \{251, 501, 1001\}$ moves $H_0^{\rm MAP}$ by $\leq 0.7\,{\rm km\,s^{-1}\,Mpc^{-1}}$, with the largest pairwise Wasserstein-1 distance between the three H_0 posteriors below $2\,{\rm km\,s^{-1}\,Mpc^{-1}}$.
- *PSD source.* TaylorF2 runs at $n_{\rm live} = 5000$ on the baseline prior with GWTC-1, `kazewong`, and BILBY reference PSDs span $H_0^{\rm MAP} = 76.4$–$77.7\,{\rm km\,s^{-1}\,Mpc^{-1}}$ and $P(H_0 > 120) = 0.043$–0.065, again with maximum pairwise $W_1 < 2\,{\rm km\,s^{-1}\,Mpc^{-1}}$.
- *Heterodyne reference.* Replacing the GWTC-1 reference parameters with the maximum-likelihood point of an initial low-cost optimisation produces IMRPhenomD_NRTidalv2 posteriors that are close to indistinguishable through the bulk of the distribution and depart only marginally in the high-H_0 tail.
- *Peculiar-velocity centre.* Sweeping $\langle v_p \rangle \in \{215, 310, 405\}\,{\rm km\,s^{-1}}$ at fixed $\sigma_{v_p} = 150\,{\rm km\,s^{-1}}$ on the IMRPhenomXAS_NRTidalv3 baseline (Figure 2b, Table 2) shifts $H_0^{\rm MAP}$ by $6\,{\rm km\,s^{-1}\,Mpc^{-1}}$ peak-to-peak across the full range (74.5 at $\langle v_p \rangle = 215\,{\rm km\,s^{-1}}$ to 68.5 at $\langle v_p \rangle = 405\,{\rm km\,s^{-1}}$), while $P(H_0 > 120)$ changes by less than 0.02. The $\langle v_p \rangle = 310\,{\rm km\,s^{-1}}$ run reproduces the IMRPhenomXAS_NRTidalv3 baseline to within 0.01 in $\ln Z$, confirming run reproducibility.
- *IMRPhenomD_NRTidalv2 companion full sweep.* The four-variant prior sensitivity sweep on IMRPhenomD_NRTidalv2 gives baseline $H_0^{\rm MAP} = 71.5\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $P(H_0 > 120) = 0.076$; flat-in-z direct $H_0^{\rm MAP} = 75.2\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $P = 0.281$; reweighted $H_0^{\rm MAP} = 74.1\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $P = 0.195$; $\sigma_{v_p} = 250\,{\rm km\,s^{-1}}$ $H_0^{\rm MAP} = 72.3\,{\rm km\,s^{-1}\,Mpc^{-1}}$, $P = 0.067$. The reweighting-capture fraction is $(0.195 - 0.076)/(0.281 - 0.076) \approx 58\,\%$, against $\approx 17\,\%$ for IMRPhenomXAS_NRTidalv3 (equation 3).

All five axes therefore produce H_0 shifts at least an order of magnitude smaller than the prior-induced shift documented in Section 4.1.